

Lab no.: 2

Study of basic network command and Network configuration commands.

Basic network and configuration commands like ping, ipconfig, nslookup, and tracert are essential tools used to diagnose and configure network connections. These commands help check connectivity, display IP configurations, resolve DNS issues, and trace the path of packets across a network.

Some basic network command and network configuration commands are as follows:

ping: Tests connectivity between your computer and another device on the network by sending packets and awaiting a response.

```
C:\Users\asus>ping google.com

Pinging google.com [2404:6800:4002:818::200e] with 32 bytes of data:
Reply from 2404:6800:4002:818::200e: time=22ms
Reply from 2404:6800:4002:818::200e: time=101ms
Reply from 2404:6800:4002:818::200e: time=130ms
Reply from 2404:6800:4002:818::200e: time=43ms

Ping statistics for 2404:6800:4002:818::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 22ms, Maximum = 130ms, Average = 74ms
```

ipconfig: Displays the current TCP/IP configuration of the computer, such as IP address, subnet mask, and default gateway.

```
C:\Users\asus>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : domain.name

Wireless LAN adapter Local Area Connection* 13:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 14:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : worldlink.com.np
    IPv6 Address. . . . . : 2400:1a00:3b26:602a:85b1:27a1:c3bb:90fb
    Temporary IPv6 Address. . . . . : 2400:1a00:3b26:602a:2421:c14b:bf2f:a2d2
    Link-local IPv6 Address . . . . . : fe80::750a:97ce:7530:bd29%19
    IPv4 Address. . . . . : 192.168.1.81
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%19
                             192.168.1.254
```

ipconfig /all: Provides detailed information on the computer's network configuration, including DNS, DHCP, and Ethernet adapter details.

```
C:\Users\asus>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-69J7G7I
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : worldlink.com.np

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : domain.name
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 10-7C-61-AD-7E-51
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 13:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #5
Physical Address. . . . . : 96-BB-43-01-DB-26
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

arp -a: Shows the IP address and MAC address mappings on the network.

```
C:\Users\asus>arp -a

Interface: 192.168.1.81 --- 0x13

Internet Address      Physical Address      Type
192.168.1.65          96-f1-ef-18-54-7c    dynamic
192.168.1.92          22-13-d5-3d-94-36    dynamic
192.168.1.96          8a-38-ae-2b-bc-af    dynamic
192.168.1.254         78-4f-24-5b-d7-7a    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static
224.0.0.252          01-00-5e-00-00-fc    static
239.255.102.18        01-00-5e-7f-66-12    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

hostname: Displays the name of your computer.

```
C:\Users\asus>hostname
DESKTOP-69J7G7I
```

netstat -a: Displays current active TCP/IP connections and listening ports.

```
C:\Users\asus>netstat -a

Active Connections

Proto Local Address           Foreign Address         State
TCP 0.0.0.0:135             DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:445             DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:5040            DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:7070            DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:7680            DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:12177           DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:49664           DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:49665           DESKTOP-69J7G7I:0      LISTENING
TCP 0.0.0.0:49666           DESKTOP-69J7G7I:0      LISTENING
```

nslookup: Diagnoses DNS problems by querying DNS servers for domain name information.

```
C:\Users\asus>nslookup google.com
Server: vip6-safenet-kmd01.wlink.com.np
Address: 2400:1a00:0:32::165

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4002:831::200e
142.250.182.142
```

pathping: A combination of ping and tracert, traces the path and measures packet loss.

```
C:\Users\asus>pathping youtube.com

Tracing route to youtube.com [2404:6800:4000:101d::80]
over a maximum of 30 hops:
 0  DESKTOP-69J7G7I [2400:1a00:3b26:602a:2421:c14b:bf2f:a2d2]
 1  2400:1a00:3b26:602a::1
 2  2400:1a00:3b02::1
 3  2400:1a00:0:1::226
 4  2400:1a00:0:37::8
 5  2400:1a00:0:41::136
 6  2400:1a00:0:41::139
 7  2400:1a00:dccc:1:72:9:128:67
 8  * * * * *
Computing statistics for 175 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
     Lost/Sent = Pct  Lost/Sent = Pct
 0      0/ 100 = 0%      0/ 100 = 0%      DESKTOP-69J7G7I [2400:1a00:3b26:602a:2421:c14b:bf2f:a2d2]
 1    10ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:3b26:602a::1
 2    22ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:3b02::1
 3    18ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:0:1::226
 4    15ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:0:37::8
 5    22ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:0:41::136
 6    18ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:0:41::139
 7    21ms      0/ 100 = 0%      0/ 100 = 0%      2400:1a00:dccc:1:72:9:128:67
Trace complete.
```

route print: route print is a Windows command used to display the IP routing table of your computer. It shows how your system decides where to send network traffic.

```
C:\Users\asus>route print
=====
Interface List
10...10 7c 61 ad 7e 51 .....Realtek PCIe GbE Family Controller
11...96 bb 43 81 db 26 .....Microsoft Wi-Fi Direct Virtual Adapter #5
18...9a bb 43 81 db 26 .....Microsoft Wi-Fi Direct Virtual Adapter #6
19...94 bb 43 81 db 26 .....Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway             Interface           Metric
0.0.0.0                    0.0.0.0          192.168.1.254       192.168.1.81        35
127.0.0.0                  255.0.0.0        On-link             127.0.0.1           331
127.0.0.1                  255.255.255.255 On-link             127.0.0.1           331
127.255.255.255            255.255.255.255 On-link             127.0.0.1           331
192.168.1.0                255.255.255.0    On-link             192.168.1.81        291
192.168.1.81               255.255.255.255 On-link             192.168.1.81        291
192.168.1.255              255.255.255.255 On-link             192.168.1.81        291
224.0.0.0                  240.0.0.0        On-link             127.0.0.1           331
224.0.0.0                  240.0.0.0        On-link             192.168.1.81        291
255.255.255.255            255.255.255.255 On-link             127.0.0.1           331
255.255.255.255            255.255.255.255 On-link             192.168.1.81        291
=====
Persistent Routes:
None
```

tracert: Displays the path packets take to reach a destination, useful for troubleshooting network routing issues.

```
C:\Users\asus>tracert google.com

Tracing route to google.com [2404:6800:4002:818::200e]
over a maximum of 30 hops:
 1  1 ms    1 ms    5 ms   2400:1a00:3b26:602a::1
 2  6 ms    4 ms   10 ms   2400:1a00:3b02::1
 3  12 ms   7 ms    5 ms   2400:1a00:0:1::226
 4  66 ms   6 ms    3 ms   2400:1a00:0:37::8
 5  6 ms    5 ms    4 ms   2400:1a00:0:41::136
 6  7 ms   63 ms   9 ms   2400:1a00:0:41::139
 7  8 ms   15 ms  12 ms   2400:1a00:dccc:1:72:9:128:67
 8  *      *      *      Request timed out.
 9  31 ms   *      67 ms   2001:4860:1:1::126a
10 197 ms  20 ms   24 ms   2001:4860:0:1::7819
11 87 ms   83 ms   30 ms   2001:4860:0:1::1687
12 49 ms   20 ms   26 ms   tzdelb-bh-in-x0e.1e100.net [2404:6800:4002:818::200e]
Trace complete.
```

Conclusion:

In this lab, we studied and executed fundamental network commands including **ping**, **ipconfig**, **nslookup**, **tracert**, **netstat**, **pathping**, and **route print**. These commands are essential for diagnosing connectivity issues, monitoring active connections, resolving DNS problems, and analyzing network routes. Through this lab, we gained practical knowledge of how these tools work together to ensure efficient network troubleshooting and management.